(1) Electromagnetic Force on Moving Charges:

$$\underbrace{\vec{E}\ ,\ \vec{B}}$$

$+q \longrightarrow \vec{v}$

Net EM force on moving charge

Lorentz force eqn.

$$\boxed{\vec{F}_{EM} = \underbrace{q\vec{E}}_{\text{electric force}} + \underbrace{q(\vec{v}\times\vec{B})}_{\text{mag force}}}$$

dir → RHTR or RHPR.

mag force → frame dependent

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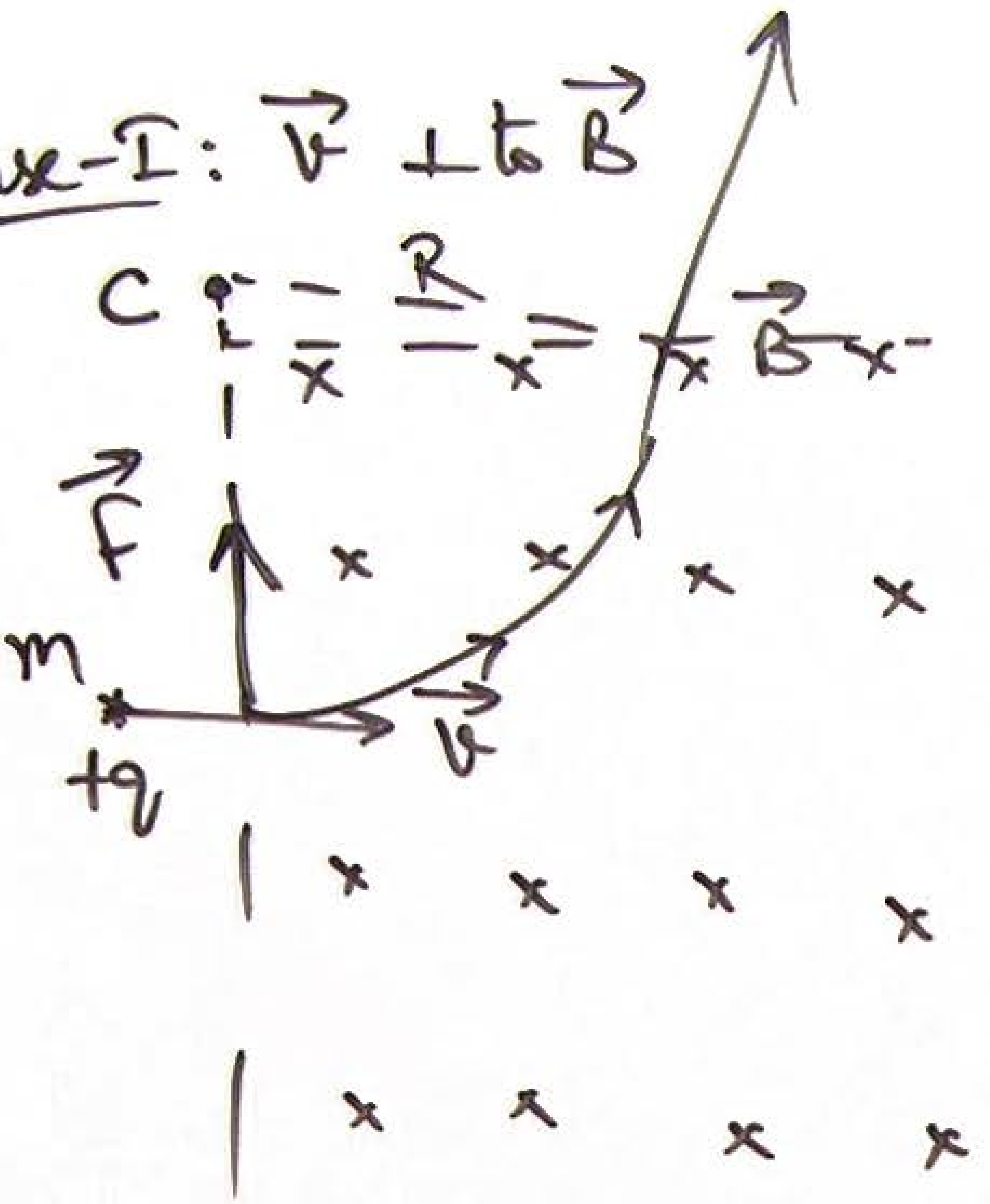
(2) Magnetic Force on Moving charges:

$\vec{F}_m = q(\vec{v} \times \vec{B})$

RHPR → finger → $\vec{B}$
thumb → $\vec{v}$ (+ve charge)
⇒ Palm face gives $\vec{F}$

Case-I: $\vec{v} \perp$ to $\vec{B}$



Magnetic force
$W = 0$
as $\vec{F}$ is ⊥ to $\vec{v}$
⇒ KE → Constant
$|\vec{v}|$ → Constant
path will be a CM.

If R is radius of UCM
⇒ $qvB = \frac{mv^2}{R}$

$$\boxed{R = \frac{mv}{qB}}$$

Ang speed $\omega = v/R = \frac{qB}{m}$ ⇒ Time period of rev

$$\left[T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}\right]$$

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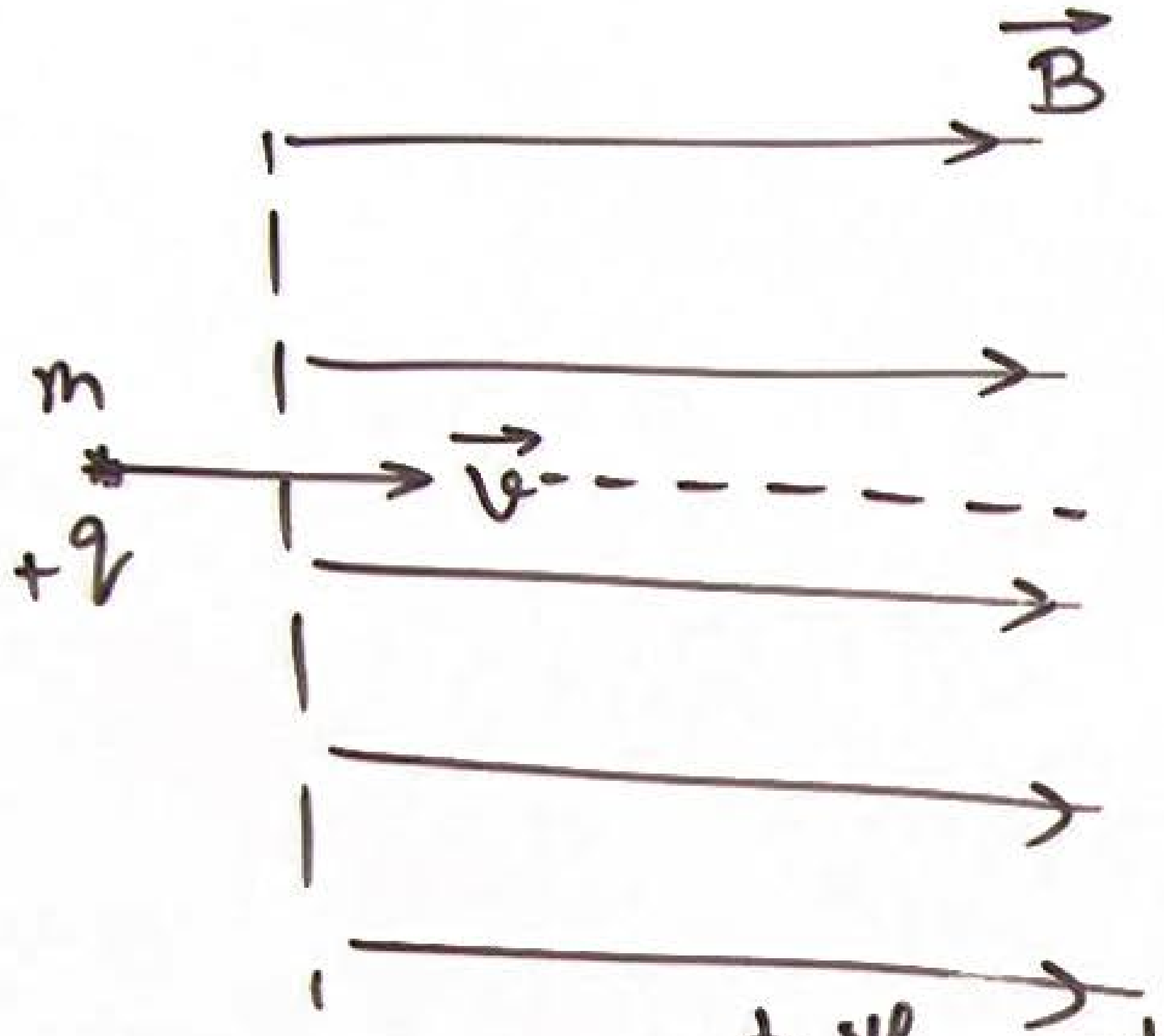
(2) Magnetic Force on Moving Charges:

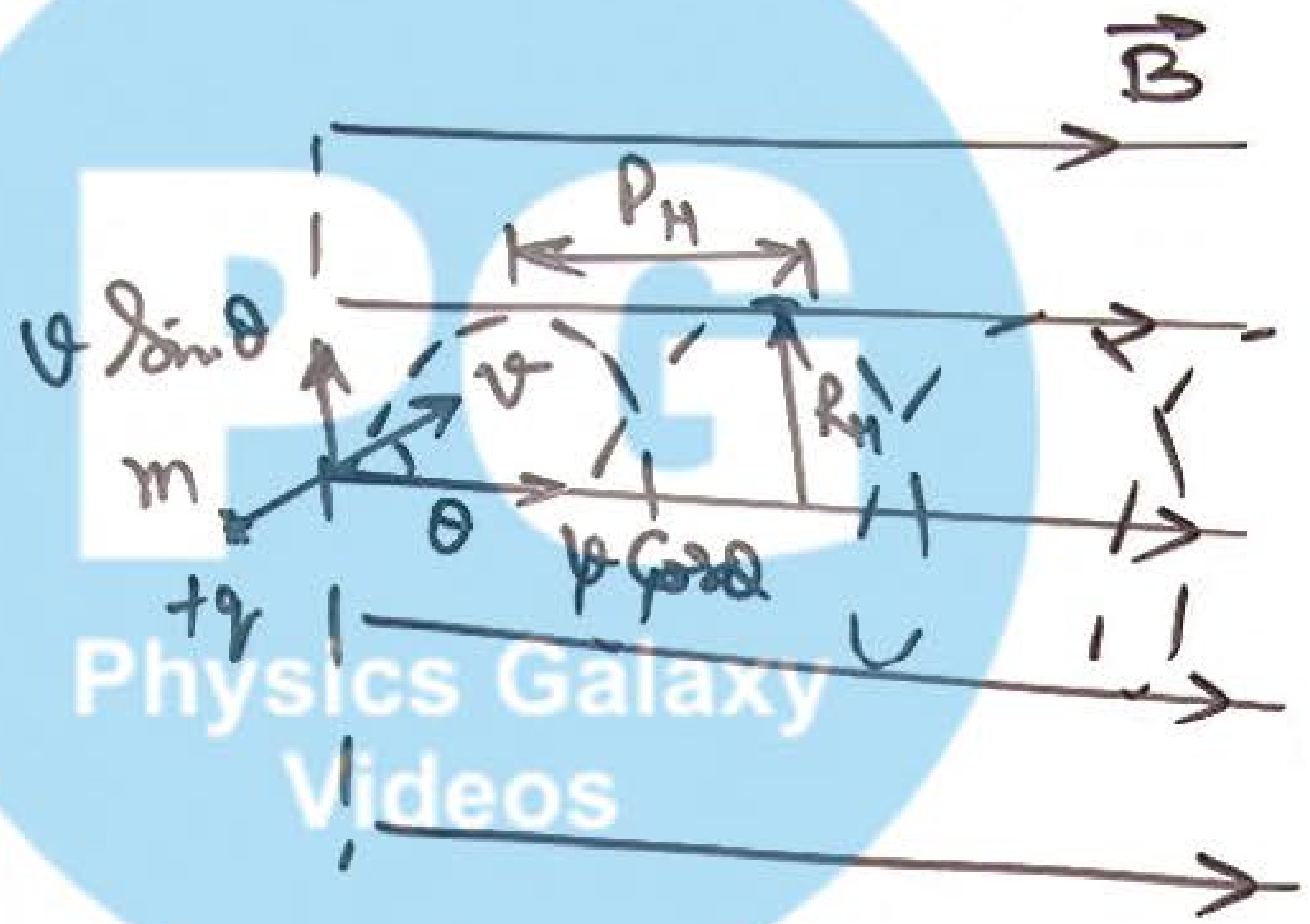
$$\vec{F}_m = q(\vec{v} \times \vec{B})$$

Case-II: $\vec{v}$ is $\parallel$ to $\vec{B}$



Path $\rightarrow$ st line

Case-III: $\vec{v}$ at angle θ to $\vec{B}$



$$F = qvB\sin\theta$$

$$T = \frac{2\pi m}{qB}$$

Radius of helix/helical path.

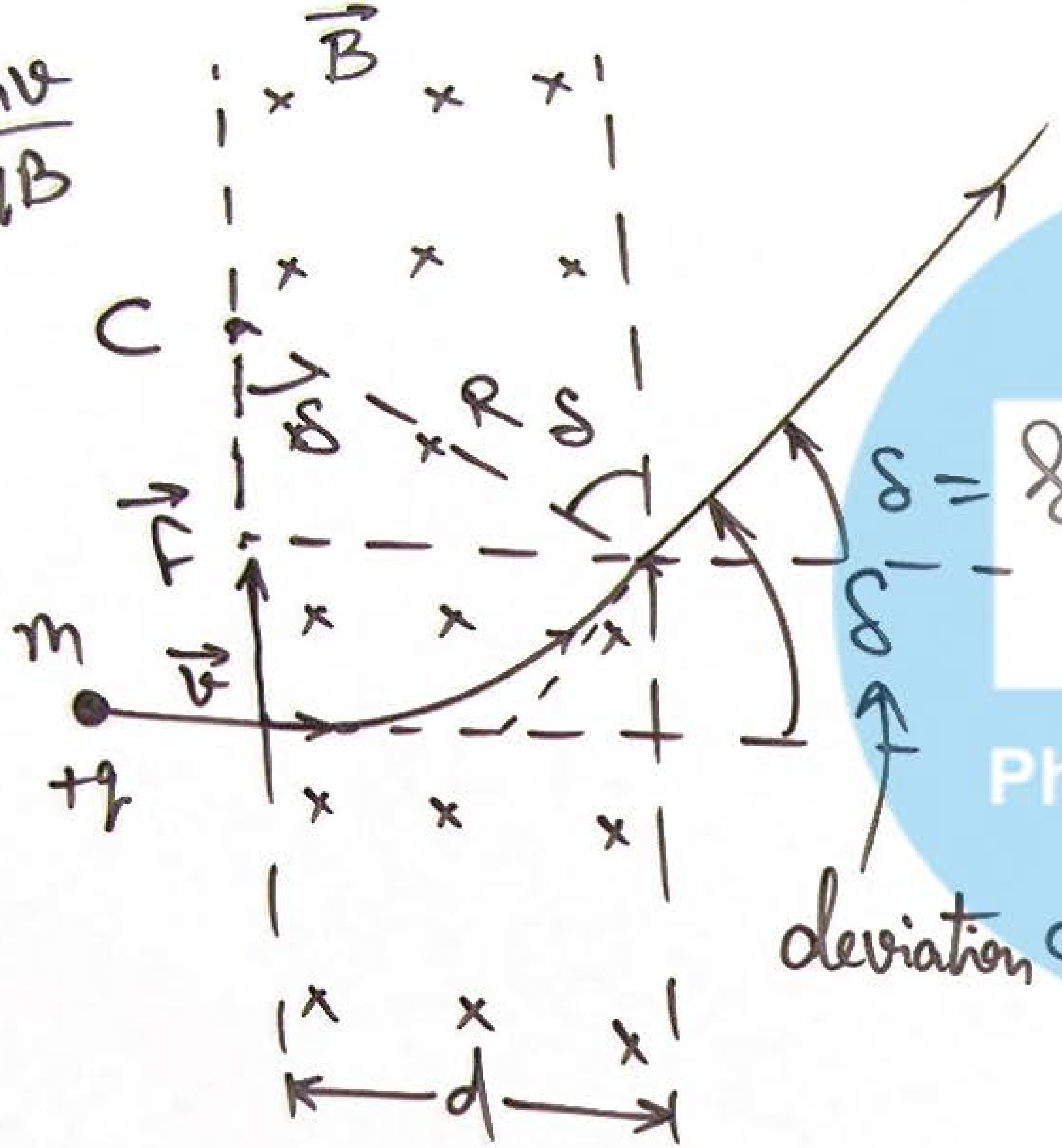
$$\boxed{R_H = \frac{mv\sin\theta}{qB}}$$

Pitch of helix.

$$\boxed{P_H = v\cos\theta \times T = \frac{2\pi m v\cos\theta}{qB}}$$

(3) Deflection by a Sector of MF:

$$R = \frac{mv}{qB}$$

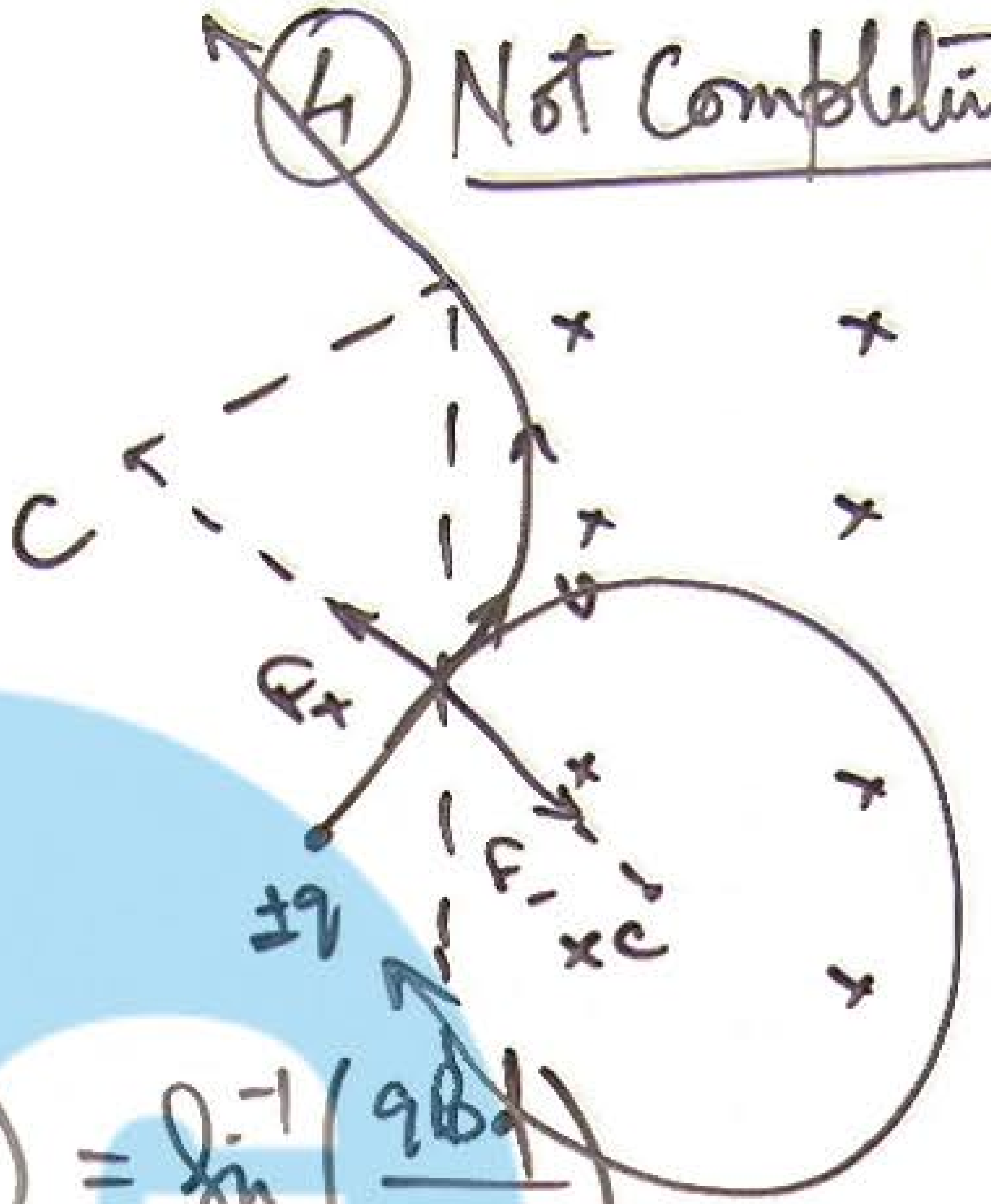


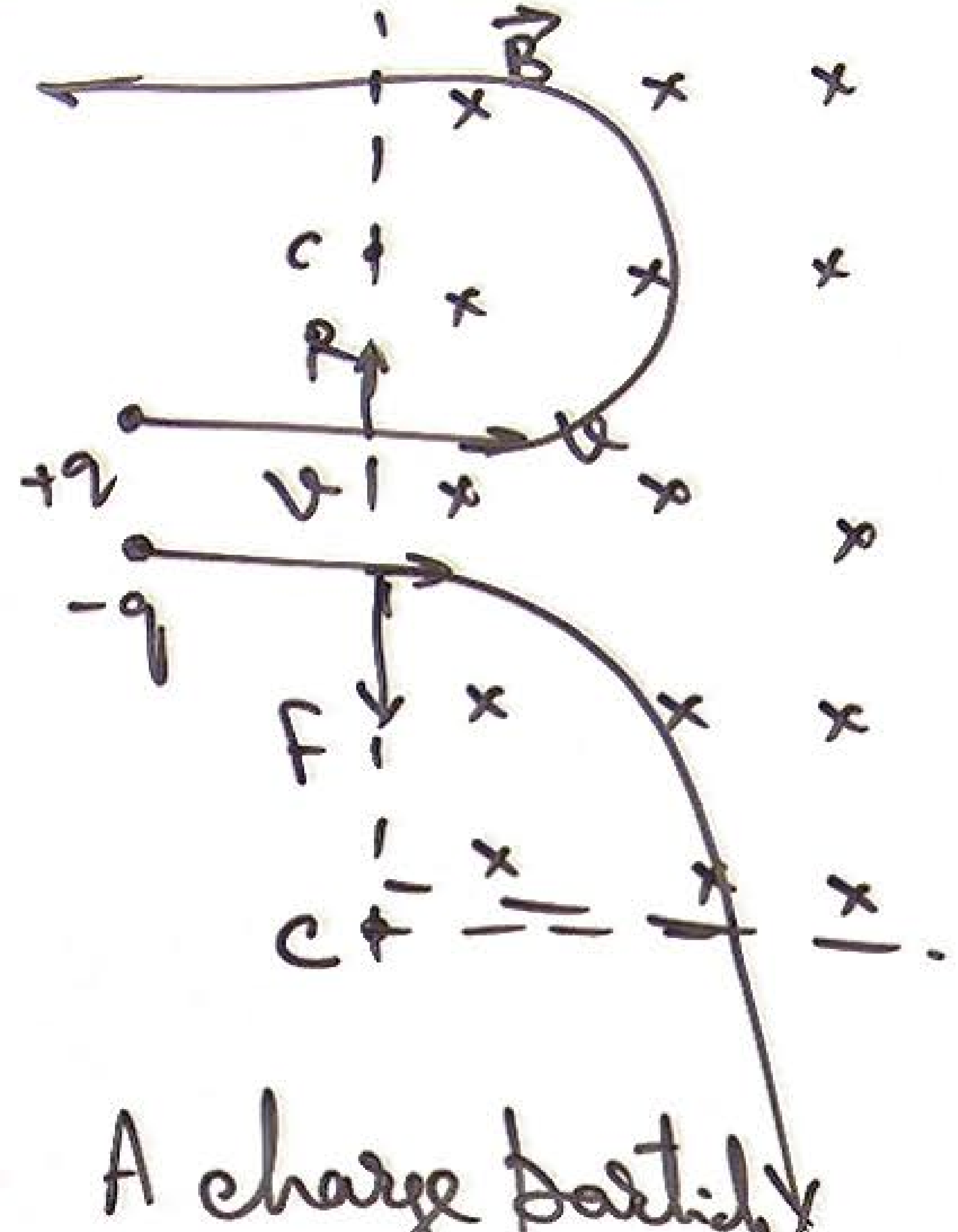
$$\delta = \sin^{-1}\left(\frac{d}{R}\right) = \sin^{-1}\left(\frac{qBd}{mv}\right)$$

deviation Angle.

if $R < d \Rightarrow \delta = \pi$

(4) Not Completing a circle inside MF:





NOTE: A charge particle can complete a circle in MF when it is projected from a pt inside of MF.

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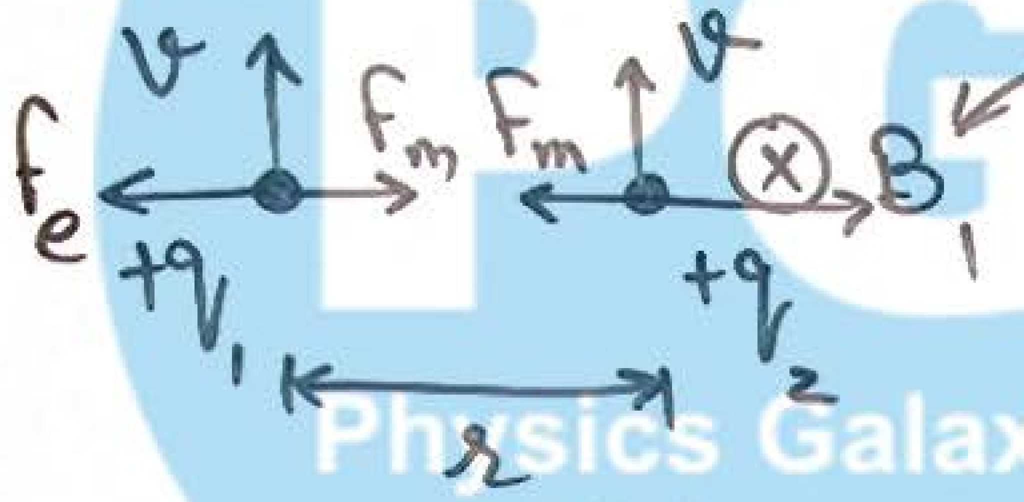
(5) Force between two parallel moving charges:

elect force betwⁿ charge $\boxed{F_e = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}}$

Ex: Can F_e balance F_m at some vel of charges?



MF due to q_1 at location of q_2

$$B_1 = \frac{\mu_0}{4\pi} \frac{q_1 v}{r^2}$$

Mag Force acting on q_2

$$\boxed{F_m = q_2 v B_1 = \frac{\mu_0}{4\pi} \frac{q_1 q_2 v^2}{r^2}}$$

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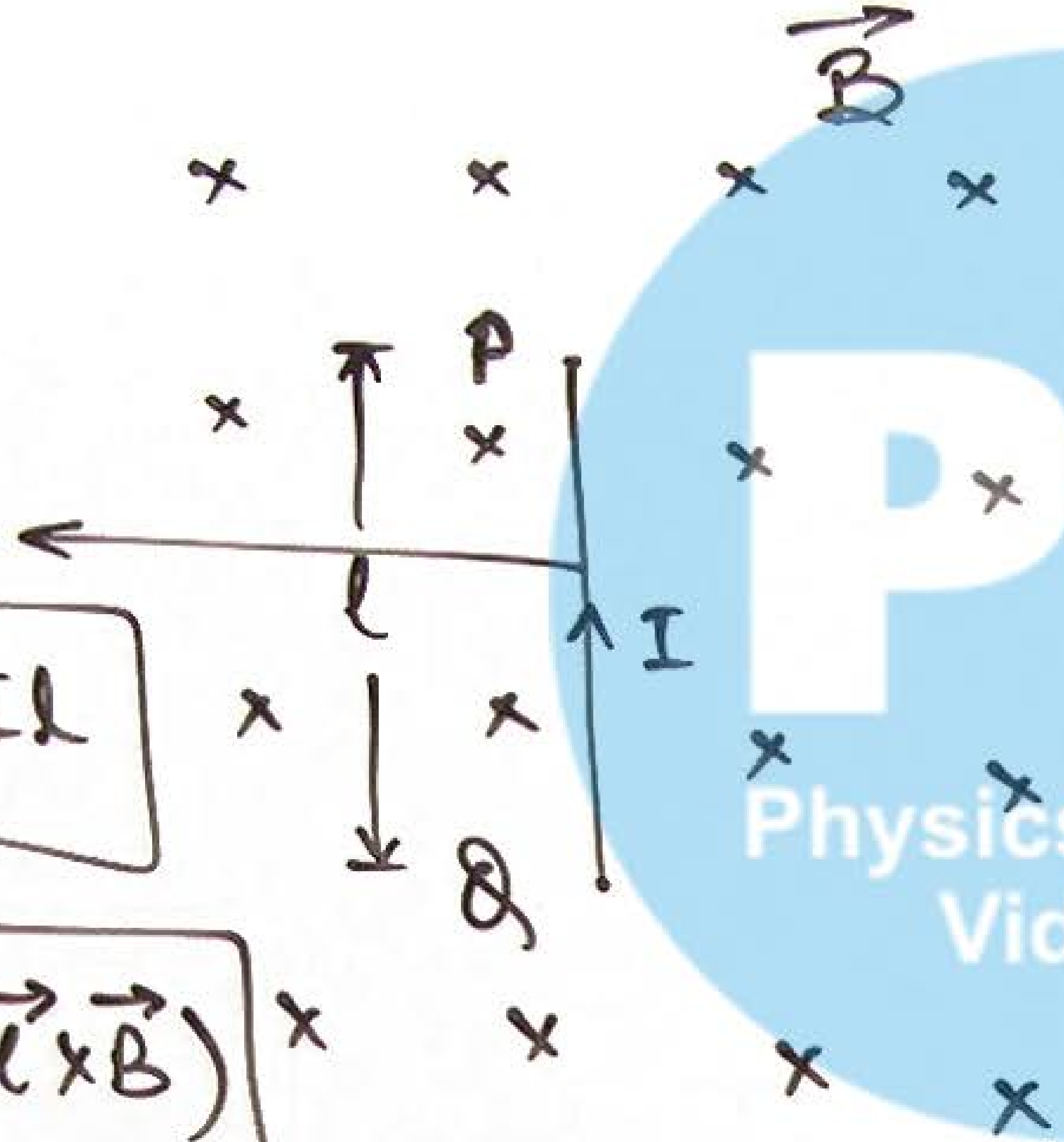
⑥ Magnetic Force on a Current Carrying Wire:



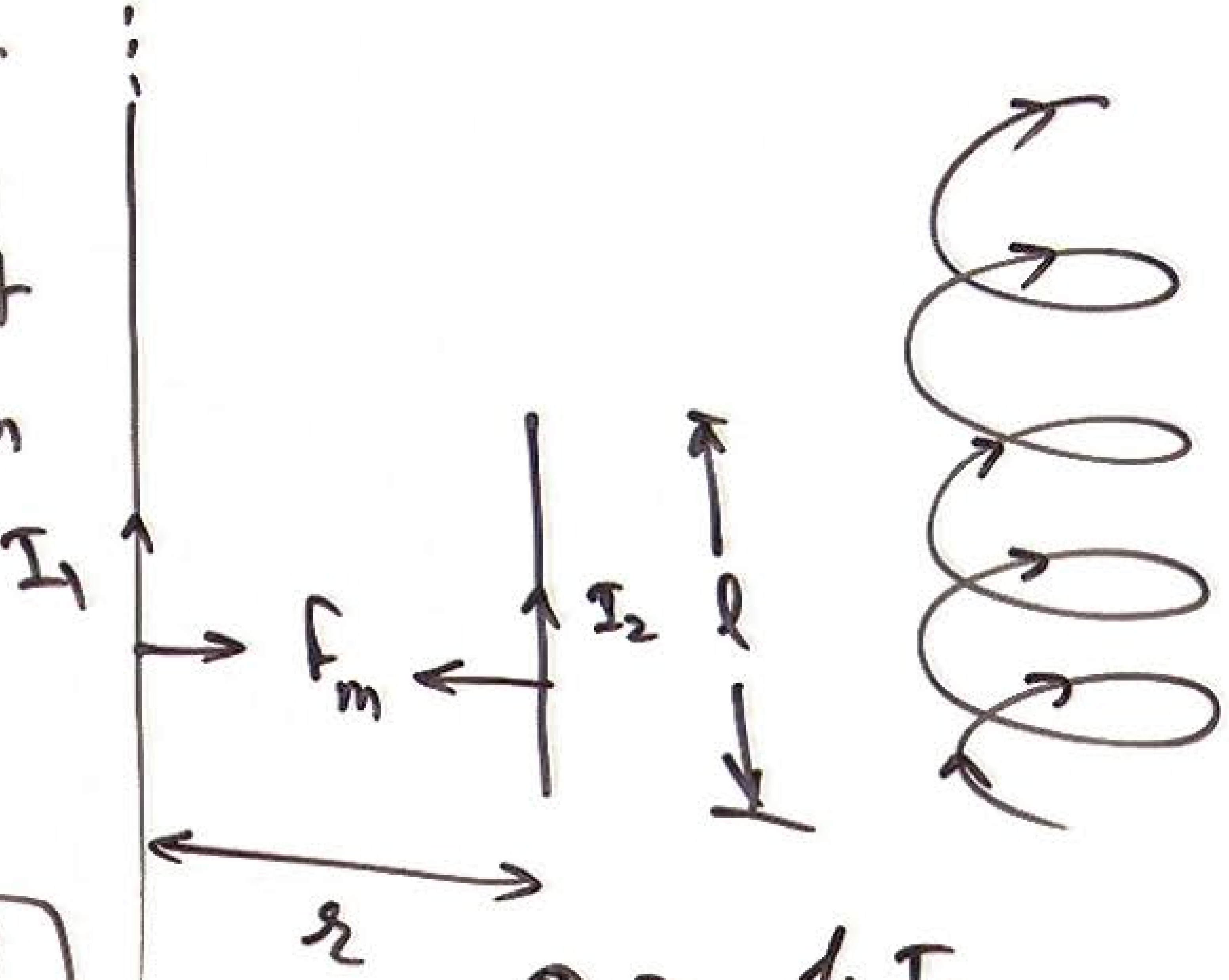
$$\boxed{F = BIl}$$

$$\boxed{\vec{F} = I(\vec{l} \times \vec{B})}$$

⑦ Force between parallel Current Carrying wire:

Same dir current ⇒ Attraction

opp dir current ⇒ Repulsion



$$F_m = B_1 I_2 l$$

$$\Rightarrow \boxed{F_m = \frac{\mu_0 I_1 I_2 l}{2\pi r}}$$

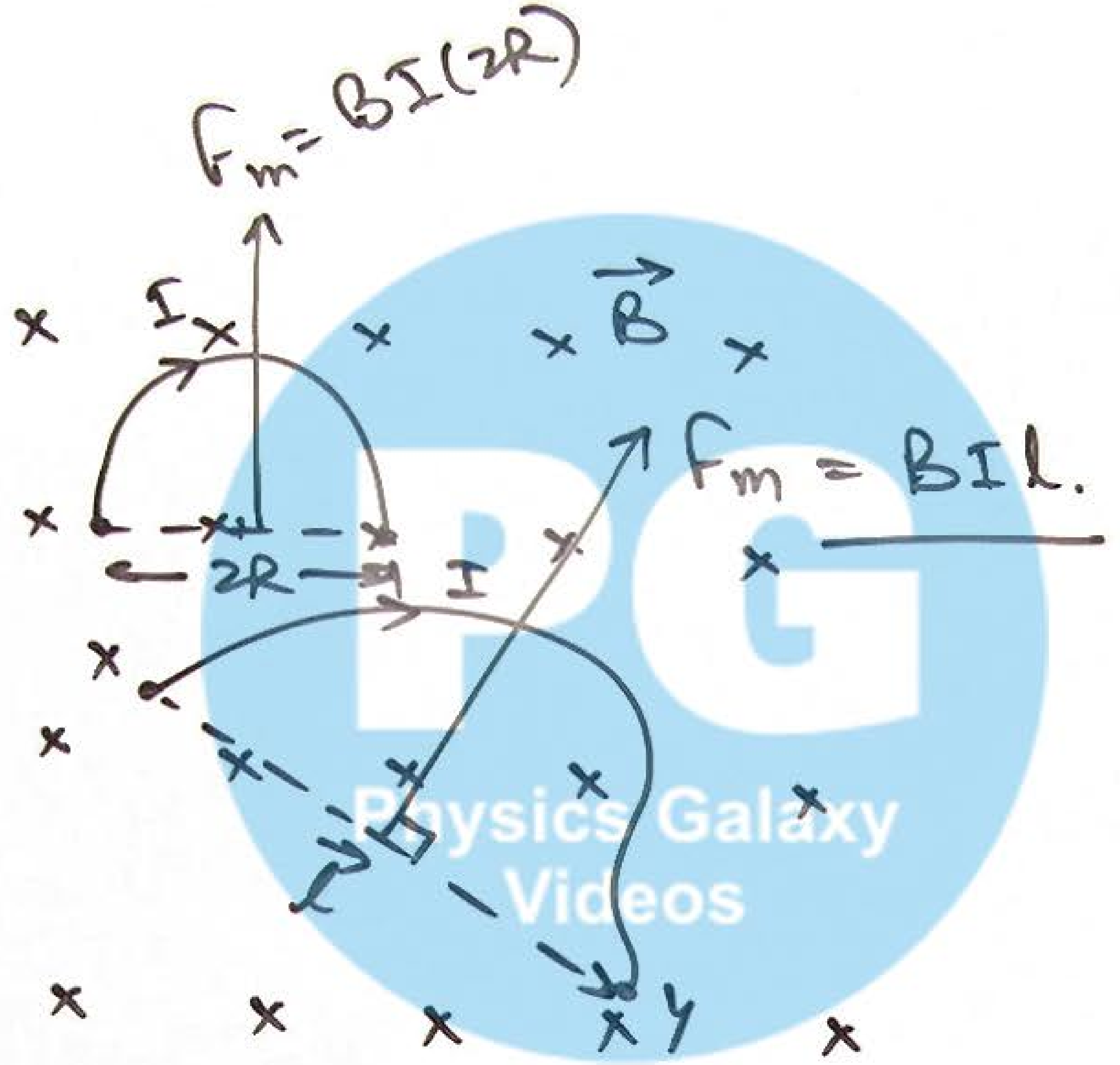
$$\otimes\ B_1 = \frac{\mu_0 I_1}{2\pi r}$$

(8) Magnetic Force on a Random Shaped wire:

$$\vec{F} = I(\vec{l} \times \vec{B})$$

↓

St distance XY



(9) Mag force on a closed Current carrying loop/coil:

$$\boxed{\vec{F}_m = 0}$$

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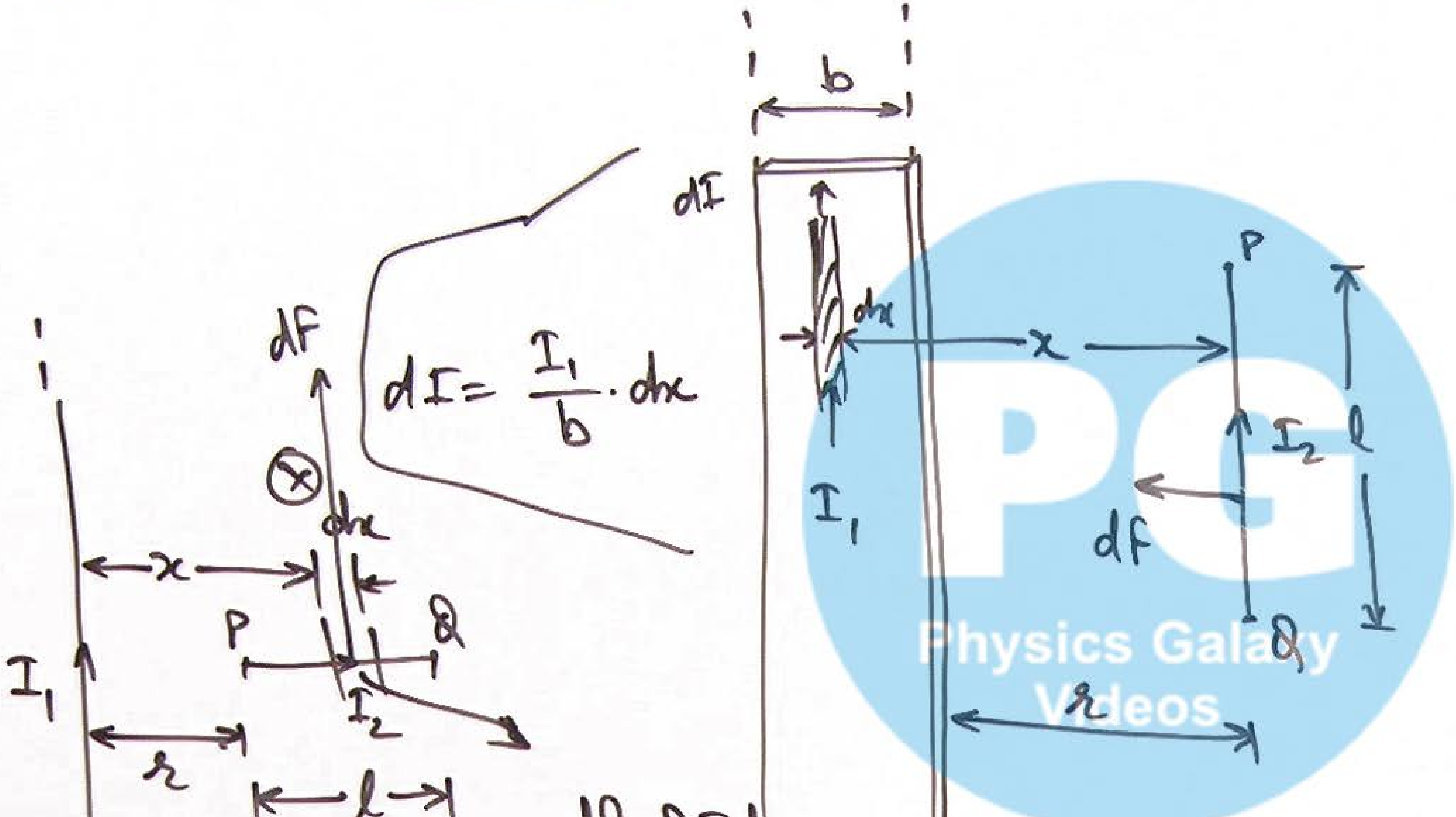
(10) Magnetic Force between a parallel wire & Strip :



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$$dF = B I_2 dx$$

$$F = \int dF = \int_r^{r+l} \frac{\mu_0 I_1}{2\pi x} \cdot I_2 dx = \text{— —}$$

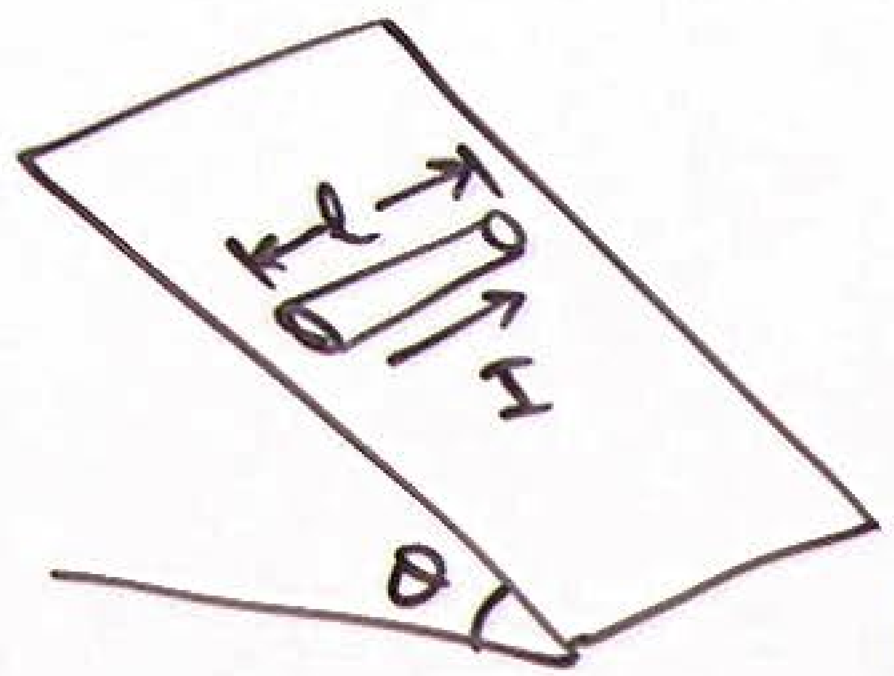
force on I_2 due to dI

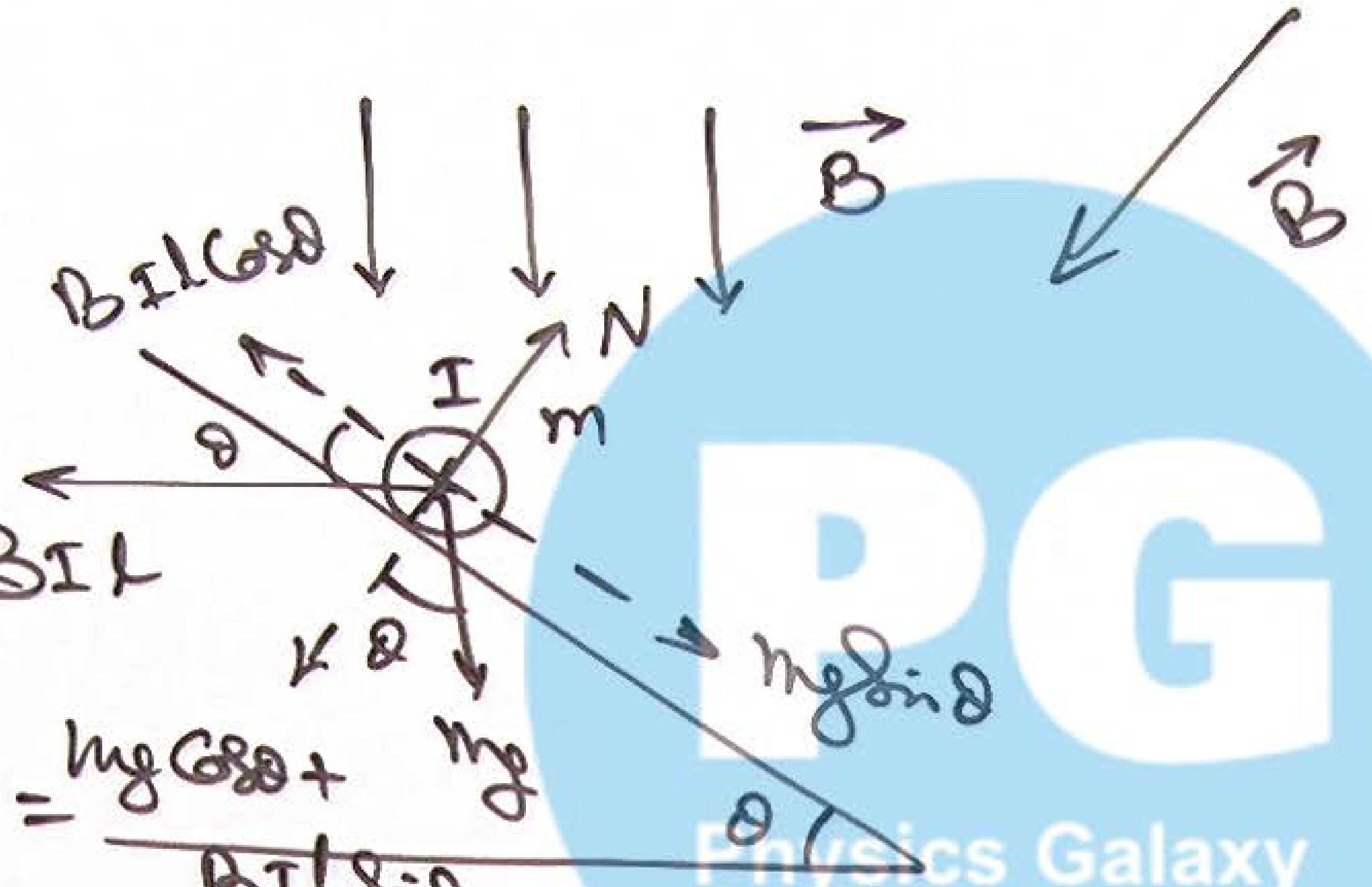
$$dF = \frac{\mu_0 I_2 dI\, l}{4\pi x}$$

$$F = \int dF = \frac{\mu_0 I_1 I_2 l}{4\pi b} \int_r^{r+b} \frac{dx}{x}$$

$$F = \frac{\mu_0 I_1 I_2 l}{4\pi b} \ln\left(\frac{r+b}{r}\right)$$

(11) Current Carrying conductor on an inclined plane:





$N = mg\cos\theta + BIl\sin\theta.$

$mg\sin\theta - BIl\cos\theta - f_r = ma$

$f_r \rightarrow \mu N$

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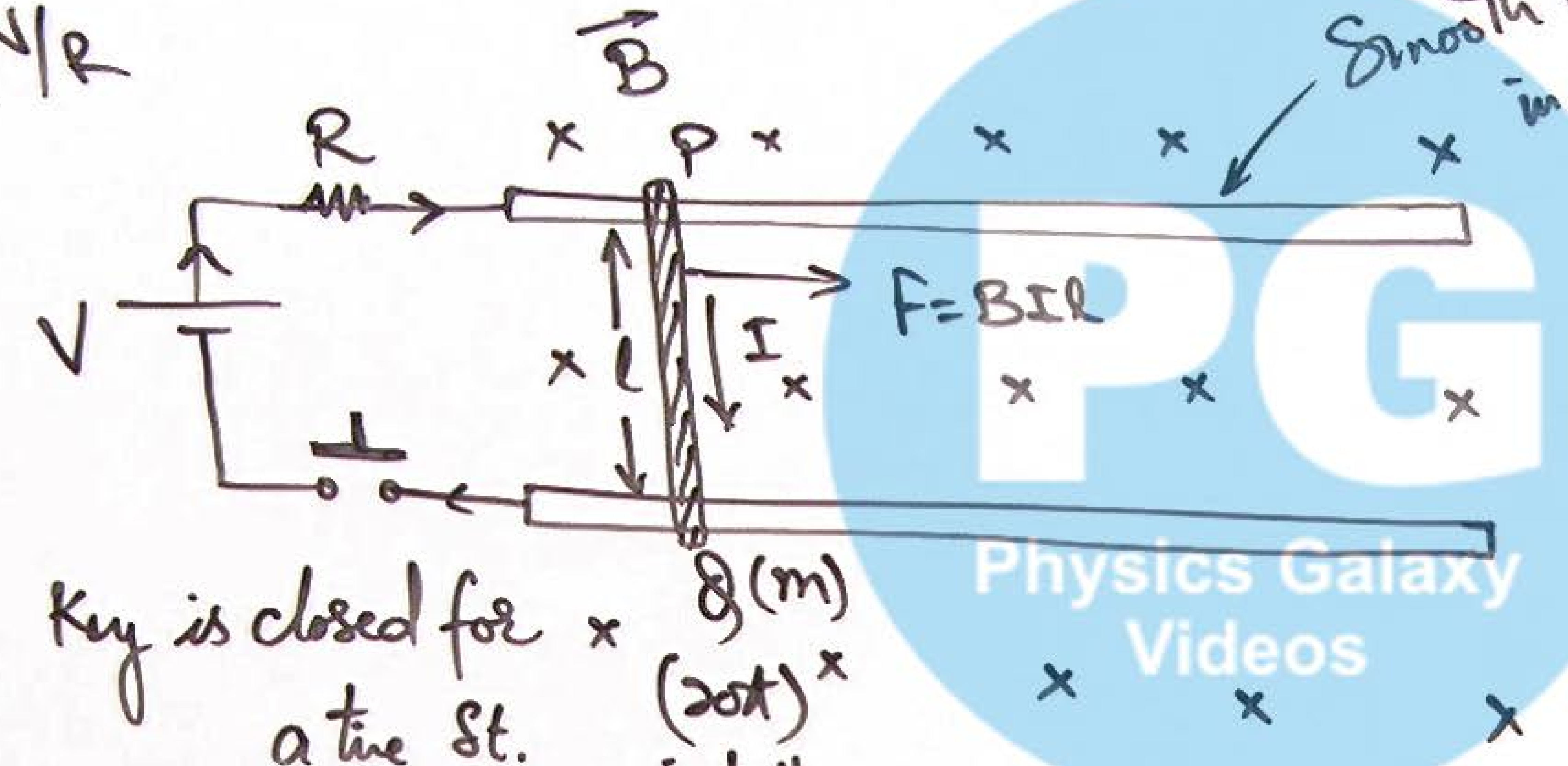
(12) Cases of Impulse by Magnetic Force:

$q = I\delta t \rightarrow$ charge passed through Conductor.

$I = V/R$



Smooth Rails in Horizontal plane

Key is closed for a time $\underline{\delta t}$.

If Conductor PQ attains a velocity v after time δt.

$$B\underline{I}l.\underline{\delta t} = mv - 0$$

$$\boxed{J = Bql} = mv$$

impulse due to magnetic force

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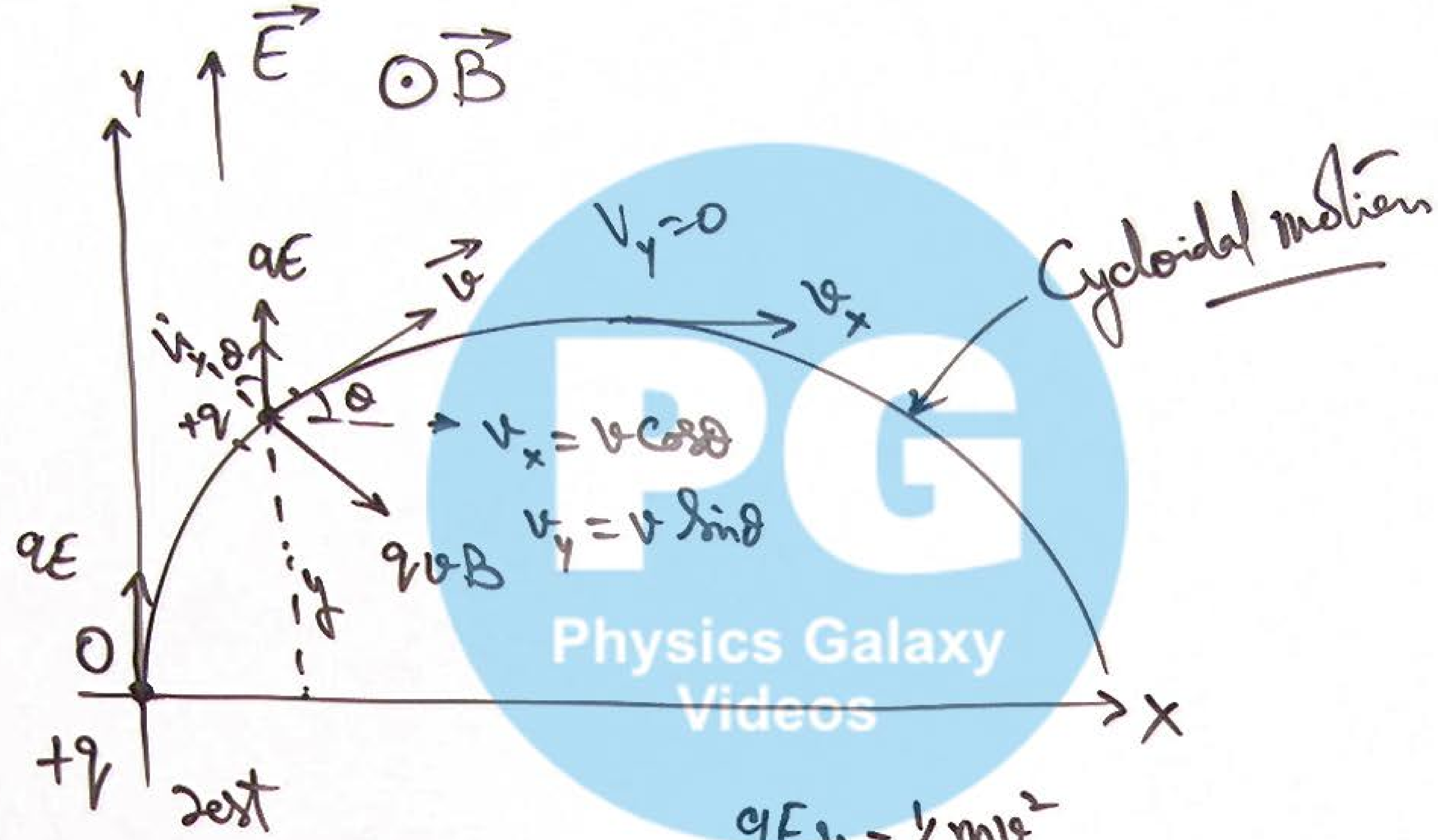
(13) Motion of a charge in Both EF and MF :

$\vec{E} = E\hat{j}$

$\vec{B} = B\hat{k}$



$qEy = \frac{1}{2}mv^2$

$qvB - qE\cos\theta = \frac{mv^2}{R}$ → radius of curvature

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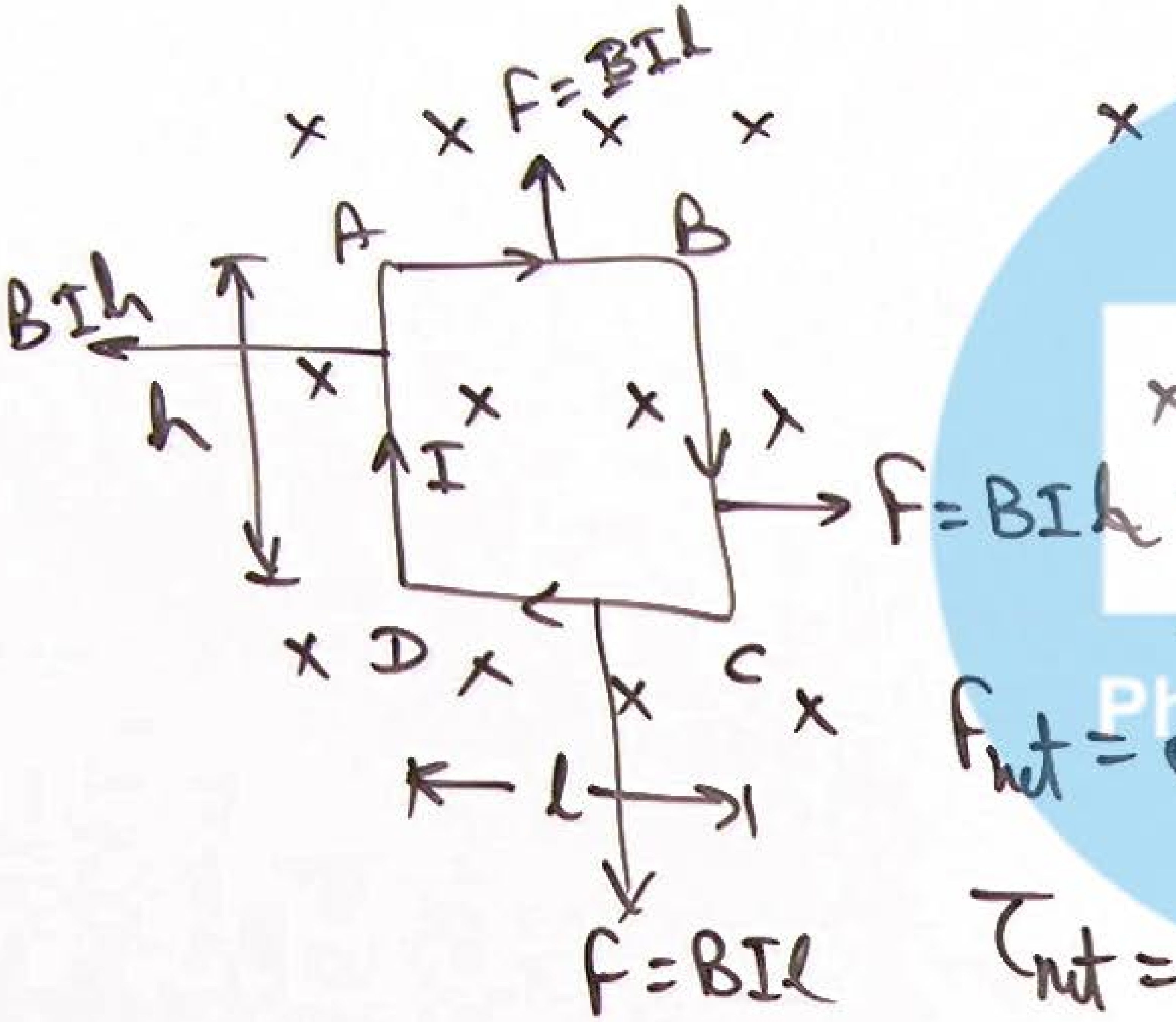
(14) Torque on a coil in MF :

torque on coil

$$\boxed{\tau = B\underbrace{IN A}\sin\theta = MB\sin\theta}$$

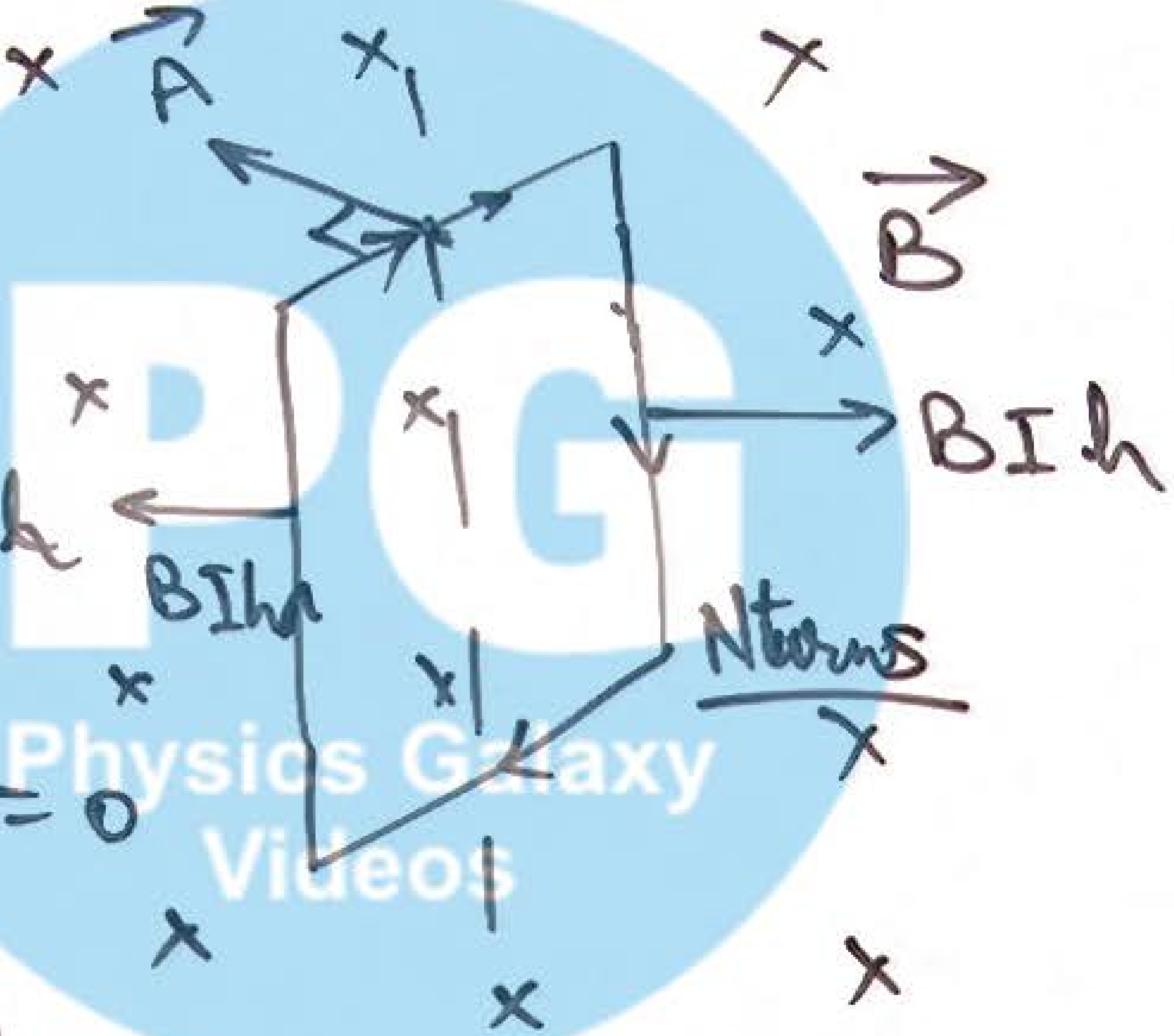
where $\vec{M} = I\vec{A}N \longrightarrow$ magnetic moment.

Vectorially $\vec{\tau} = \vec{M} \times \vec{B}$



$F_{net} = 0$

$\tau_{net} = 0$



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(15) Interaction Energy of Coil in MF & its equilibrium :

"Potn Energy"

$$\boxed{U = -\vec{M}\cdot\vec{B} = -MB\cos\theta}$$

Similar to PE of electric dipole

$$U = -\vec{P}\cdot\vec{E}$$

unstable eqm

$$U_{max} = MB \quad (\text{at } \theta = 180^\circ)$$

Stable eqm.

$$U_{min} = -MB \quad (\text{at } \theta = 0^\circ)$$

W/D in rotation/changing orientation of a coil in MF :

$$W_{ext} = U_f - U_i$$

[Initial angle betn $\vec{M}$ & $\vec{B}$ is θ_1 & final angle is θ_2]

$$\boxed{W_{ext} = MB(\cos\theta_1 - \cos\theta_2)}$$

or $W_{ext} = \int_{\theta_1}^{\theta_2} \tau \cdot d\theta$

ext.

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(16) Magnetic Flux through a Current Carrying Coil :

$\phi = \vec{B}\cdot\vec{S}$ if $\vec{B}$ is uniform

$\phi = \int \vec{B}\cdot d\vec{S}$ if $\vec{B}$ is Non uniform

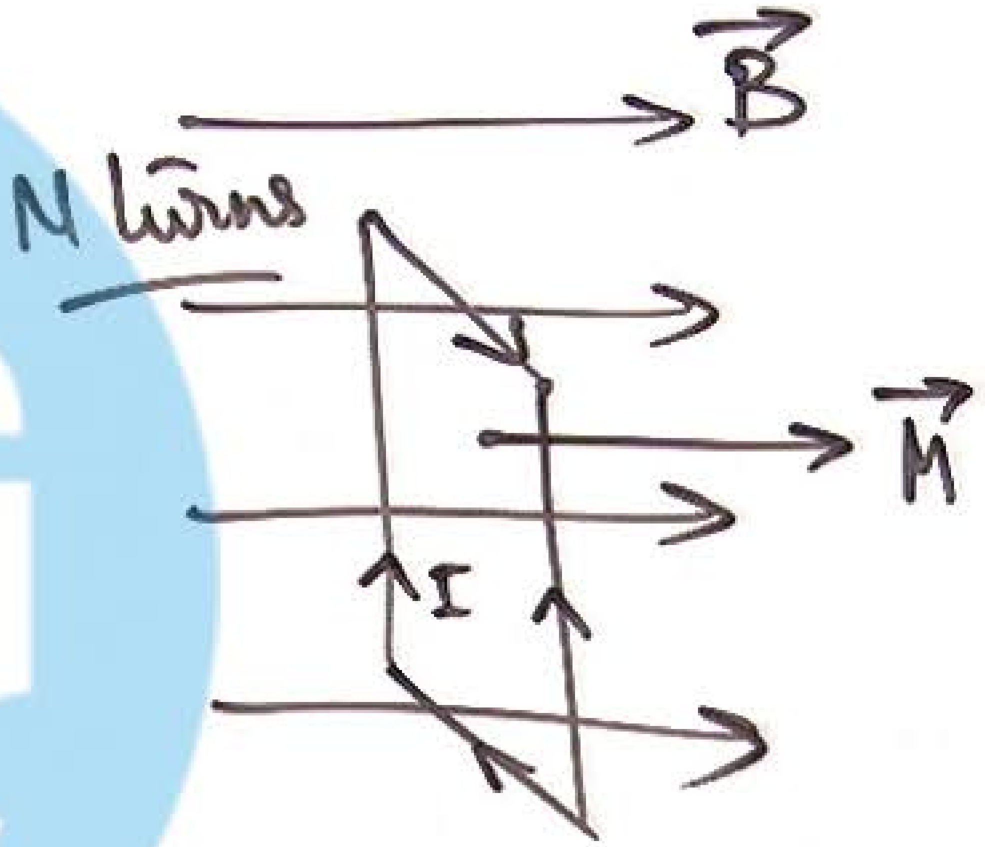
$$\boxed{\oint_M \vec{B}\cdot d\vec{S} = 0}$$ V. imp.

WD in rot^n of a coil in MF

$W = INAB(\cos\theta_1 - \cos\theta_2) = I[\phi_{1L} - \phi_{2L}]$

$$\boxed{W = -I\Delta\phi}$$

where $\Delta\phi = \phi_f - \phi_i$



passing flux $= \phi$

linked flux $= N\phi$

$\phi = \vec{B}\cdot\vec{A} = BA$ (ckwise)

$\phi = -BA$ (ackwise)

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(17) Moving Coil Galvanometer :

On passing current in coil torque on it is

$\tau = MB \sin 90^\circ$

restoring torque

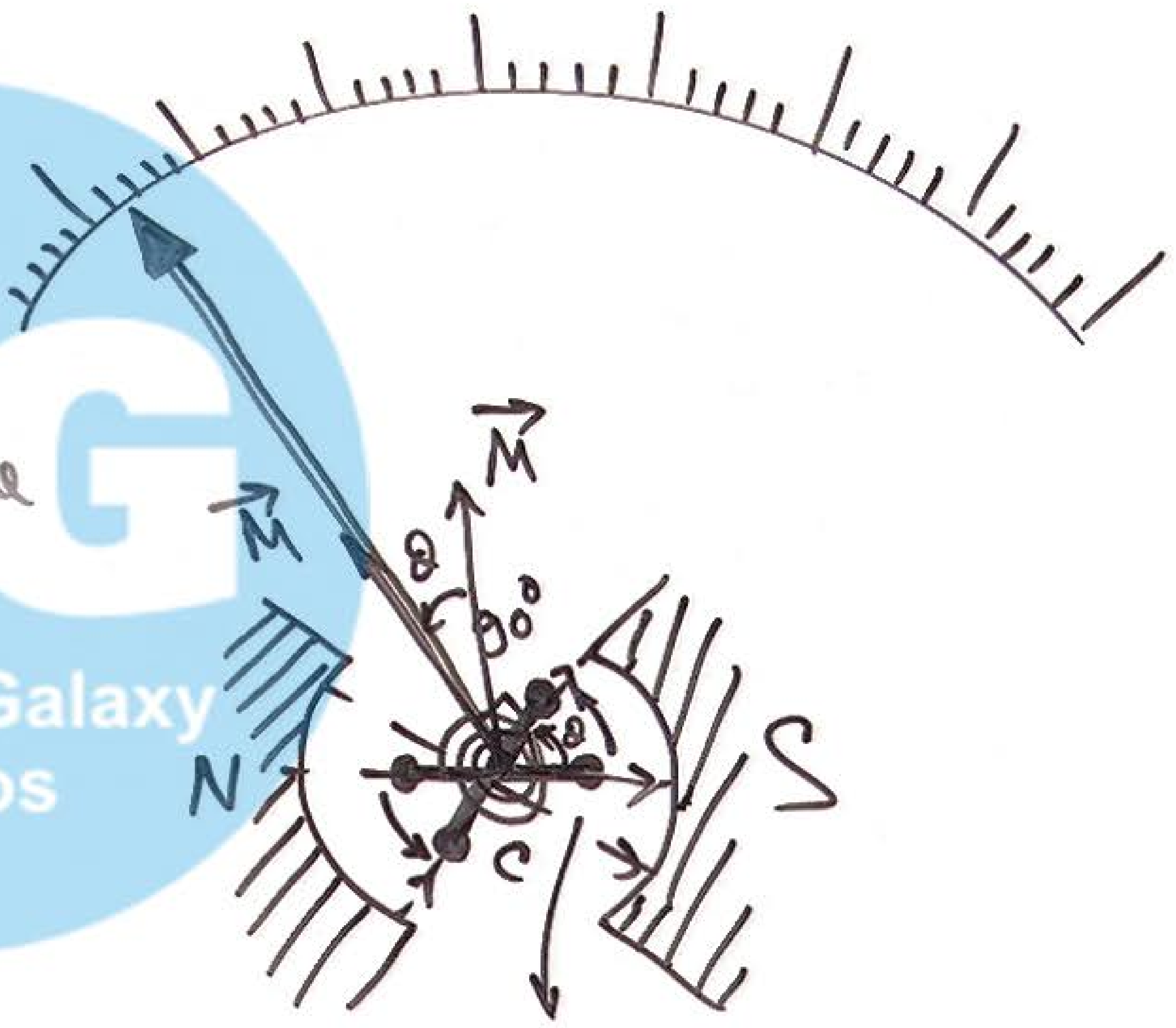
$\tau = INAB = C\theta$

"Current Sensitivity"

$\theta = \left[\frac{NAB}{C}\right] I \longleftarrow I = \left(\frac{C}{NAB}\right)\theta$

$\left[\frac{NAB}{C}\right]$ → rad/Amp.

$\left(\frac{C}{NAB}\right)$ → Amp/rad.



$M = INA$

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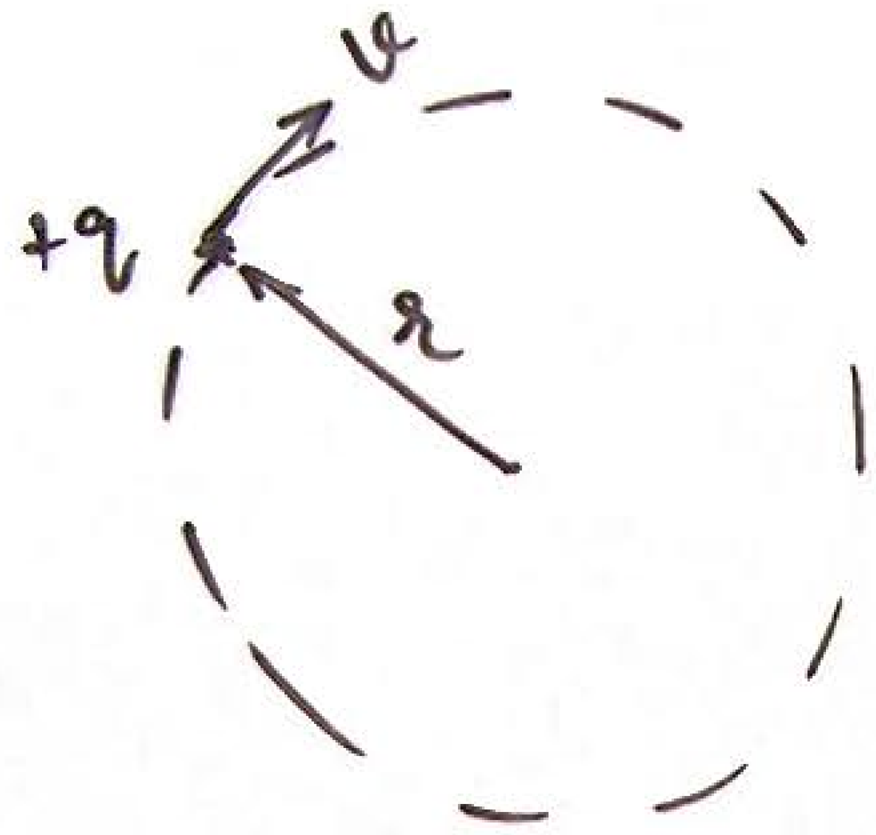
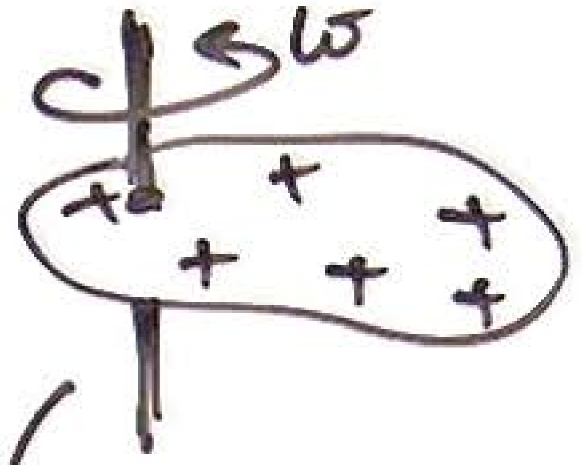
(18) Relation in Magnetic Moment & Angular momentum of a Rotating charge:

$$\boxed{\frac{M}{L} = \frac{q}{2m}}$$

Valid for uniformly charged & uniformly dense bodies.



$I = \frac{q\omega}{2\pi} \rightarrow$ Magnetic Moment of charge $M = I\pi R^2$



Angular Momentum

$L = I\omega$

$M = \frac{q}{2m} \cdot L$

Magnetic moment $M = \frac{q}{2m} \cdot I\omega$

$= \frac{q}{2\cancel{m}} \cdot \frac{2}{3}\cancel{m}R^2 \cdot \omega$

$= \frac{1}{3} q\omega R^2$ ✓

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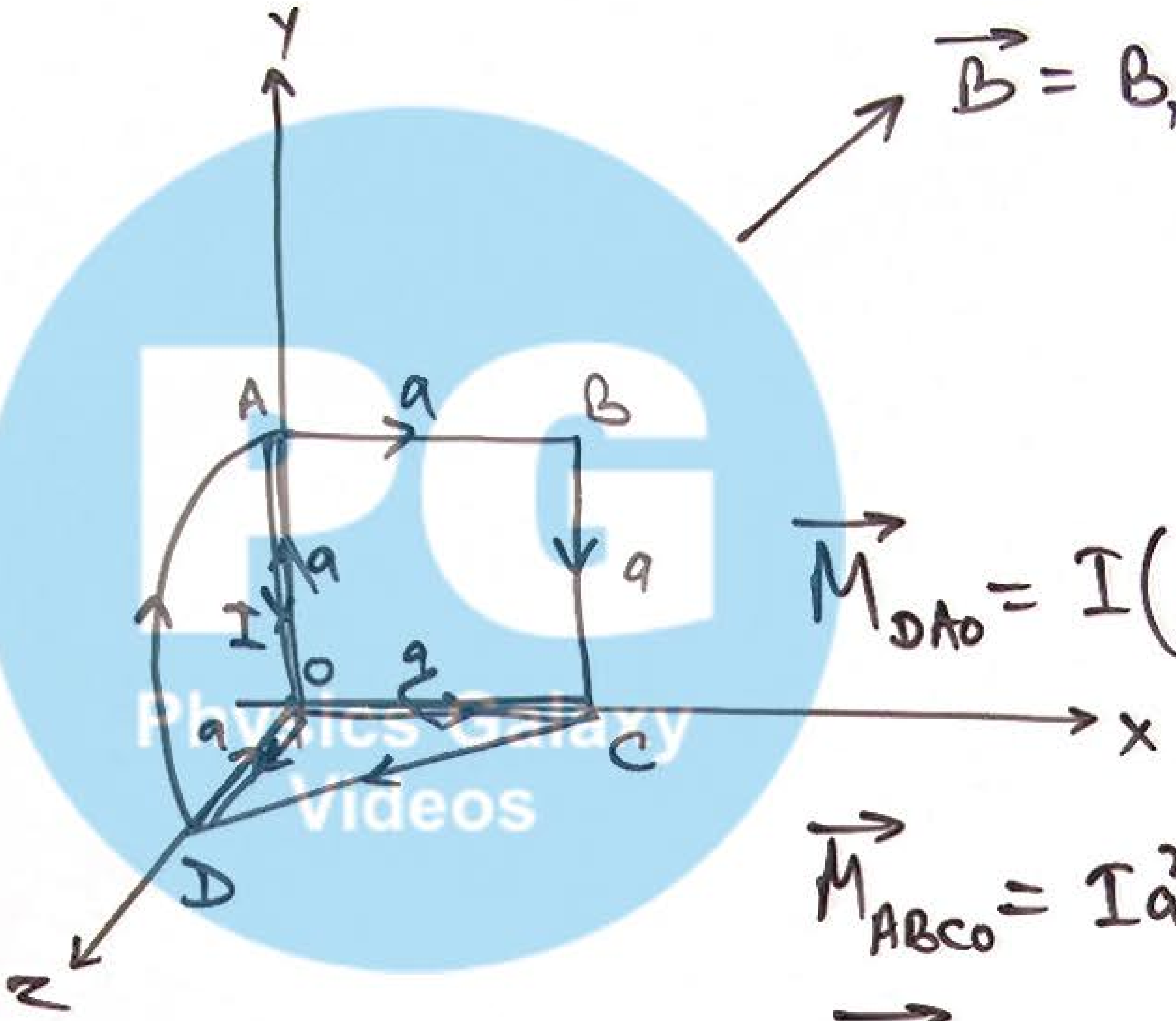
(19) Torque on a 3D loop in Magnetic Field:

$$\boxed{\vec{\tau} = \vec{M} \times \vec{B}} \checkmark$$

$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

$$\vec{M} = \vec{M}_{ABCO} + \vec{M}_{CDO} + \vec{M}_{DAO}$$



$$\vec{M}_{DAO} = I\left(\frac{\pi a^2}{4}\right)(-\hat{i})$$

$$\vec{M}_{ABCO} = Ia^2(-\hat{k})$$

$$\vec{M}_{CDO} = I\left(\tfrac{1}{2}a^2\right)(-\hat{j})$$

NOTE: A freely placed loop starts rotating about an AOR which passes through its COM & along the dir of $\vec{\tau}$.

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